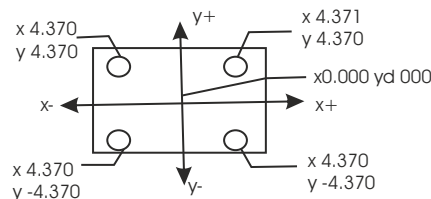


- Q.28 What is ATC Why it is used in CNC machine.
- Q.29 What is pallet system, explain any one of the pallet system.
- Q.30 Give the importance of role of computer in automation.
- Q.31 Write the main advantages of LVDT.
- Q.32 What is axis identification, how it is used in CNC machine.
- Q.33 Write the main guidelines to be followed in the management of a tool room.
- Q.34 Give the function of magnetic tape in CNC machine.
- Q.35 Difference between absolute and incremental coordinate system.

#### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain canned cycle with an example.
- Q.37 What are the input mediums that can be used to feed data or program in CNC machine? Write the advantage and disadvantage of each.
- Q.38 Write a complete drill program for the following part. Assume any missing data of dimension.



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#### 5th Semester/ Mechatronics Subject : CNC Machines & Automation

Time : 3 Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Full form of MCU is  
a) Machine computer unit  
b) Machine control unit  
c) Machine control universal  
d) Machine computer universal
- Q.2 Choose the Robert component from the following  
a) Microcomputer b) Coaxial cable  
c) Arm d) Software
- Q.3 Which of the following code will be gives circular interpolation clockwise  
a) G00 b) G01  
c) G92 d) G02
- Q.4 The function of slide ways in CNC is to \_\_\_\_\_.  
a) Reduce friction b) Reduce wear  
c) Improve smoothness d) All of above

(20)

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- Q.5 LVDT has got \_\_\_\_\_ number of windings  
 a) 04                                      b) 02  
 c) 03                                      d) 08
- Q.6 The CNC control system which has no feedback are also known as.  
 a) Uncontrolled systems      b) Controlled systems  
 c) Closed system                  d) Open system
- Q.7 Which of the following code is used in coolant on.  
 a) M07                                      b) M03  
 c) M04                                      d) M01
- Q.8 \_\_\_\_\_ is not type of NC program reader.  
 a) Card reader                      b) Punched type reader  
 c) Laser type reader              d) None of them
- Q.9 The purpose of using Re-circulating ball screw nut mechanism in CNC machine is \_\_\_\_\_.  
 a) To reduce the setup time  
 b) For higher surface finish  
 c) For carrying out up milling  
 d) To remove backlash
- Q.10 Which of the following Feed Back device translate physical motion in to electrical data.  
 a) Encoder  
 b) Transducer  
 c) Digital system monitoring  
 d) None of the above

### SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Expand PLC.  
 Q.12 Define Slide ways.  
 Q.13 Name any two type of NC programming reader.  
 Q.14 \_\_\_\_\_ code is used for Dwell function.  
 Q.15 Give binary equivalent of 27.  
 Q.16 G-codes are also known as.  
 Q.17 What is part programming.  
 Q.18 What is the function of Actuators.  
 Q.19 Write the application of DNC machine.  
 Q.20 Define Group Technology.

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write the main advantages and disadvantages of DNC machine.  
 Q.22 What is PLC control? Write the function of PLC.  
 Q.23 Define the function of rollers and coating in CNC machine.  
 Q.24 Difference between transducers and sensors.  
 Q.25 Explain briefly on-line diagnostic.  
 Q.26 Write any six major electronic component problem.  
 Q.27 Write a short note on CAD and CAM.